Examiner reports

Q1.

This question discriminated well between students of varying abilities. The majority of students understood that they had to integrate and chose the appropriate technique, often applying integration by parts successfully. A significant number of students made no attempt to go any further and completely ignored the constant of integration. These students generally achieved three marks.

Students who realised they had to determine the constant of integration often made good progress and scored 7 or 8 marks. Those who dropped marks at this point generally showed insufficient detail to fully justify their answer. Several very thorough, complete solutions were seen.

Q2.

Part (a) was done well by the majority of students. Most knew how to set up the initial differential equation and those who did generally scored full marks. There were a few numerical slips seen and some students did not complete the argument to show the required result.

A high proportion of students made good progress on part (b) with over half of students scoring 2 or 3 marks. Having correctly integrated, many students gave no consideration to the constant of integration and, although it was zero, its value needed to be found to correctly show the given result.

Part (c)(i) was well attempted and saw the full range of marks awarded. Over half of the students scored at least 4 marks with 6/6 being the most frequent score awarded. Many students realised that simultaneous equations had to be solved and usually eliminated a variable to form a correct equation. Students often made algebraic slips when solving their equation and a more efficient use of allowed calculator functions would have helped here.

In part (c)(ii) many students made some good criticisms of the equations they had found, but did not link these to the model used by the trader, which was the rate of sales is proportional to $\frac{8-t}{x}$.

Q4.

A full spread of marks was seen on this question, with about 20% of the entry gaining full marks, and about another 20% only scoring 2 marks or fewer.

Part (a). Many students attempted this integral by using parts and produced some complicated looking expressions using incorrect integrals. This integral cannot be done by parts; some students did realise this and swapped to the required substitution method. Those who did use an appropriate substitution were usually successful, with an occasional error being made in the coefficient. Few students could do the integral by inspection.

Part (b). Most students knew they had to separate the variables and integrate both sides, with the vast majority using their result from part (a). Most used correct notation in the separation, but some students read e^{2y} as e^2y and some did not multiply across correctly by e^{2y}. Of those who did, most solved the integral of e^{2y} correctly. Those who also solved the integral in x correctly usually went on to complete successfully, although the common error was to think $\ln(a + b) = \ln a + \ln b$. Students who did used their result from part (a), whatever it was, could score method marks for using (1,0) to find a value for their constant, and to use logarithms to express their final answer as $y = f(x)$.

Most students did include a constant, and did attempt to take logs to obtain a final solution.

Q5.

There was a full spread of marks seen in this question with about a quarter of the students scoring 8 or more of the 10 marks available, but a similar proportion scoring 2 or less.

(a) Most students recognised that this required integration by parts, although some just 'integrated' the t term and the cosine term independently, for no credit. Some tried to use substitution, and indeed some students substituted $u = \frac{\pi t}{4}$ and then used parts. Most identified the parts the right way round but those who did not should have found the integral becoming impossibly harder but they were often determined to get a result, again for no credit. Of those who started correctly, many made a mistake in writing down the parts formula, often confusing the sine and cosine terms. Many had the required coefficient $\frac{4}{\pi}$ inverted. Many who had the coefficients correct at this stage then made an error when integrating the sine term, often getting $\frac{8}{\pi}$ or similar. Many made a mistake when simplifying a correct expression, which was not penalised until part (b).

(b) Most students separated the variables correctly, and usually used the correct notation. Occasionally the x was dropped or the dx or dt did not appear or was in the wrong place. A few attempted to 'integrate' the expression as given in the question to produce answers of no value. Most gained the mark for integrating x correctly, although some used logarithms due to incorrect separation. Most of the students also gained the method mark for using their result from part (a) in an attempt to find a constant of integration. Some students just substituted $t = 0$, $x = 4$ into the differential equation itself, which made no sense. Those who had the correct form of the integral from part (a) could then score the remaining method mark for calculating firstly x^2 and then x . Many were correct to this point and then misunderstood the request to give their answer to the nearest centimetre or they evaluated using degrees rather than radians. About 8% of the entry gave a fully correct solution.

Q6.

A substantial proportion of candidates solved this differential equation successfully, which is an impressive achievement given that there is a considerable amount of careful manipulation involved. Most candidates started with a correct separation, with most of these using correct notation. Many integrated $\frac{1}{y^2}$ correctly, but this was not the case with the integration of $x\sin 3x$. Some showed no detail of an attempt at parts and scored no further marks. Of those who did show stages in their parts integration, some made sign and / or coefficient errors. Most candidates knew that they were to include a constant and attempted to find it, but they needed to have a $\cos 3x$ and a $\sin 3x$ term in their expression to demonstrate knowledge of sin $\frac{\pi}{2}$ = 1 and cos $\frac{\pi}{2}$ = 0 when finding the value of the constant. The final stage of factoring out a 9 and inverting was done successfully by most who had got this far, although some incorrectly added the constant after inverting and scored no further marks. Those who failed to gain the last mark usually made an error in their last line or two of working, making a sign error or dropping a 3 or even the x.

Q7.

For part (a), some student wrote down the correct differential equation apparently fully understanding all the information given and interpreting it correctly. However, all sorts of errors abounded in other attempts, some not even involving a derivative, and some with derivatives in x and y. Some attempts bore little or no relation to the information given, although credit was given if ± (2 – h) appeared in an otherwise incorrect expression. Many had a spurious t and / or h, either as a multiple or power, and the k appeared in a variety of places. Some students did not even form an equation, leaving a proportionality sign in their answer.

In part (b), most students knew they were expected to separate the variables and did it correctly, although there were some notation errors in the positioning of dx, at the front rather than the rear of the integrand. Those who failed to separate the variables, just produced nonsense.

Most students integrated k dt successfully, but relatively few made good progress with the integral in x . No guidance was given, but students split fairly evenly between attempts by parts and by substitution, with similar success, or lack of it, via either method. Many students just integrated each term in the product; presumably this was an attempt at parts. Those who showed the parts and how the integral might be achieved scored a method mark, but many could not integrate $\sqrt{(2x-1)}$ correctly, making an error in the required coefficient.

Those who used substitution usually chose to put u = 2x – 1 but did not always change their whole integral to be in terms of u; a missing du was penalised for the method mark. Some with a correct first stage then did not multiply out to obtain an integrable form, but just integrated each term in the product. All students, no matter what their attempt at the integral, could obtain a method mark if they included a constant and tried to find it using the given initial conditions. Those few who did integrate and find the constant correctly, usually went on to calculate the final requested value of t correctly.

Q8.

A full range of marks were seen in this question. Some candidates had little or no idea as to how to solve a differential equation, whereas some gave concise and correct solutions and could use their solution to solve the problem in the given context.

In part (a)(i) most candidates separated the variables correctly, although some poor notation was seen and some mishandled the $\sqrt{x}$. Many also made errors in the integration, often getting the coefficient wrong in one or both integrals. Although many did add an arbitrary constant, many also failed to complete the question as requested and by writing x in terms of t. Of those that did, many failed to square their expression in t correctly; $(f(t) + C)^2 = (f(t))^2 + C^2$ was a common error. Many apparently didn't review their

solution when given the form of answer in part (a)(ii).
When manipulating an expression containing a constant some, for example, renamed

$\frac{C}{2}$ as C

and sometimes then confused themselves; whereas others correctly introduced a new letter for the revised constant.

In (a)(ii) many went on to find a correct value for the constant and give the required form, as they used a previous line of working from part (a)(i) rather than their final answer. Any attempt to find a constant using the information given was credited, but many tried to force their solution into the form of the answer given, rather than review their work for an error.

In part (b)(i) the question confused a lot of candidates and many assumed the greatest height occurred at cos bt = 0 and not −1. Many candidates didn't attempt this part.

In (b)(ii) for those who set up a correct equation, the most common error here was to use degrees instead of radians; so the more common answer was 207.3, it apparently not occurring to candidates that a time of over 3 minutes was rather long for a fairground car to reach a height of 5 m. Many did not set up a correct form of the equation and so could gain no marks.

Q9.

In part (a) most candidates at least used $\frac{dA}{dt}$ in their attempt at a differential equation although

some did use other letters. Very few equated to a constant of either sign. The term "constant rate" as given in the question just wasn't understood, with most taking the rate of change to be proportional to the surface area, or to time or functions of these such as an exponential function. Only about 5% of the candidates set up the correct equation, which meant they could achieve little in part (b)(i).

In part (b)(i) those who had set up a correct equation in either $\pm k$ usually completed correctly, using integration and the conditions given. Some did attempt to integrate their differential equation, and this was give credit; many did get as far as a value for k, in which algebraic and numerical errors were condoned for the method mark, but somewhat less than 15% of the candidates did get this far.
In (b)(ii) most candidates were able to interpret the given solution in part (b)(i) to answer this correctly with over 60% of the candidates gaining this 1 mark.

Q10.

In part (a) a repeated term partial fractions expression such as this is the most complex form in the specification, but most candidates approached it with confidence and demonstrated they knew how to find the coefficients. Most chose to use values of x to find C and A in that order, and then either a third value of x, or equated coefficients in order to find a value for B. This approach was generally more successful than those who equated coefficients to set up and solve three simultaneous equations. Most had the value of C correct, and similarly A although

$\frac{1}{4}A = 1$ often resulted in $A = \frac{1}{4}$. However, about 60 % of the candidates had all of A, B

and C
correct.

In (b) many candidates ignored the partial fractions they had just found, and attempted all sorts of nonsensical ways of "integrating" the right hand side, some before even attempting to separate the variables, using what appeared to be an attempt to use parts. Some who did separate the y term to the left hand side, then attempted to multiply out the right hand side and integrate term by term, taking no notice of denominators. About 40% of the candidates gained no credit in part (b). However, many did know the first step in solving the differential equation was to separate the variables and use the partial fractions. The notation was often poor as was the algebra in handling the 2 $\sqrt{y}$, the latter often appearing as such on the left hand side and not as its reciprocal. Many mistakes were made in attempting to integrate the various terms. Indices caused problems for many with the term $\frac{1}{\sqrt{y}}$ being expressed as both $y^{\frac{1}{2}}$ and $y^{-\frac{1}{2}}$. Credit was given for integrating $\frac{k}{\sqrt{y}}$ to $2k\sqrt{y}$, for any k. In the two log integrals on the right-hand side coefficient and sign errors were common and many thought the third term had a logarithmic integral as well. Others assumed this term integrated to $\frac{x}{1-x}$ as given in the answer. Some candidates did integrate this term correctly by inspection, whereas others used substitution, also correctly. Most candidates included a constant and tried to find it using the boundary conditions given, but they were only credited with the correct value if was exact, and found from a correct expression. The penultimate mark was for correctly combining the logarithm terms in a candidate's solution; many failed to do this correctly, multiplying terms when they should have been dividing or vice versa depending on signs. Those who had made errors and so didn't have all the coefficients equal to 2, did make some valiant efforts using powers to combine their terms, and got the credit if it was done correctly. Everything had to be correct to gain the last mark, and this was achieved by rather less than 5% of the candidates.

Q11.

Marks for this question covered the whole range from 0 to 6. Many candidates knew the topic well and gave a concise and correct solution for full marks. Most candidates separated the variables correctly with conventional notation, the common error being to have $\frac{1}{y}$ on the left hand side followed by a log integral; there were also many errors in integrating Although most candidates knew this involved sine, there were errors with signs and the coefficient was often given as $\frac{1}{3}$ rather than 3. Some candidates dropped the 3 completely.

Candidates who had only made sign or coefficient errors could gain partial credit when finding a constant of integration and many did. Pleasingly, virtually all candidates who attempted this worked in radians. Some candidates omitted the constant altogether and substituted the given values into their solution, or even the given differential equation to produce nonsense. Some candidates confused themselves by multiplying by 2 before finding the value of their constant and then halved it again at the end.

Q12.

(a) Attempts at this question varied from no attempt, through superficial attempts to some fully correct solutions. Most could separate the variables correctly, although a few did attempt "integrals" of f(x)dt. Some candidates inverted the given differential equation and so could attempt direct integration, which was a quite acceptable method. Most integrated kdt correctly, but the integral of $(x+1)^{-\frac{1}{2}}$ defeated many. Often there was an error in the power, before or after integration, and also in the coefficient. However, most candidates did include a constant and attempted to find it from the given initial conditions, for which they gained credit. However, some did some algebraic manipulation before adding a constant, and so could gain no credit, as this was then incorrect. Many candidates who had correctly integrated the expressions failed to gain the last mark, which was for giving the answer in the form x = f(t), because they couldn't square their expression correctly, often omitting the "middle" term.

(b) Candidates who had interpreted the context correctly could gain at least one mark here by substituting t = 60 into their result from part (a) and solving for x. However, many confused the given t = 60 with the given initial condition x = 80. Relatively few correct answers were seen, with many candidates obtaining results of over 100% or indeed negative results. These were not apparently queried, with little evidence of checking for errors.

(c) (i) Very few candidates wrote down a correct differential equation. Some who had seemingly understood the context, wrote down the correct expression in A, but did not include the constant of proportionality. Many candidates mixed up the variables with x, and less so y, appearing, although most had a "differential equation" of some sort. Some candidates even included units in their differential equation, which was condoned if at the end of a correct expression.

(ii) For many candidates this was the only part of this question, they attempted. Candidates who had understood the context and started with A = 4.5 could mostly solve the equation for t successfully. The relatively rare error was in the algebraic manipulation or in attempts to take logarithms, such as log(1 + 4e$^{-0.09t}$) = log1 – 0.09log 4t or similar. Some made an error in miscopying –0.09.

Those candidates who commonly started with A = 50 or A = 0.5 encountered the need to take the logarithm of a negative number in their working or obtained a negative value for t, which most then chose to ignore. They gained no credit. Some did realise their error in interpreting the context, and reverted to the correct value for A.

Q13.

The full range of marks was seen in this question. Some candidates showed no understanding of how to proceed to solve the differential equation or how to use the given result to answer part (b). In contrast, some gave a concise and accurate derivation of the solution and understood how to use it in part (b). In part (a), algebraic error in the exponential term was common in attempted rearrangements, although some candidates did not separate x and t before attempting integration.

Similarly, in the integration of the exponential term, the sign or coefficient was often incorrect, although most candidates could integrate kt correctly. In part (a)(ii), most candidates had included a constant, and of these many changed its value whilst rearranging, by dividing by 2 or changing signs without saying so. Due to sign errors, many could not take the logarithm of both sides correctly; although taking logarithms of a negative number was condoned for the method mark, ln $(-kt^2)$ becoming $-$ln (kt^2) or similar was not. There was a lot of fudging of signs and coefficients in attempts to get to the given answer.

In part (b), candidates who went awry did not seem to realise they had been given the solution for x, and that the given 6000 matched the initial conditions given in part (a). Many seemed to think that the 6000 was $\frac{dx}{dt}$, and that the given 10 was t, and they solved for x. Those who did write down the correct expression usually evaluated it correctly, with a few overlooking the request for the nearest hundred, this being a modelling situation. Relatively fewer candidates were successful in part (b)(ii). Some who had realised they were to solve for t, with x set to zero, did not deal with the 2 in the expression before exponentiation. A few candidates, who were otherwise correct, overlooked taking the square root in finding t.

Q14.

Not all candidates appeared to have time to attempt this question and often solutions appeared rushed rather than thought out. However many of those who did attempt the question solved the differential equation in part (a) correctly. Most candidates separated the variables correctly using conventional notation, but then there were often errors in the integration. Common ones included the omission of a constant, incorrect integration of both x and cos2t or use of degrees instead of radians when evaluating the constant. Some candidates just substituted the given conditions straight into the differential equation, producing nonsense.

In part (b)(i) some candidates substituted $\pi/4$ into dx/dt instead of their solution from part (a). Another error was to substitute $13\pi/4$ for t. There were, however, some correct solutions showing understanding and most candidates who attempted this did show they intended to work with sin 26, even if they were trying to use it in an incorrect expression or with their calculator in degrees. Most gave answers to the accuracy required by the question.

In part (b)(ii) those who had successfully solved the differential equation in part (a) usually earned the method mark here. However correct answers were rare, as many candidates arrived correctly at a negative value for 2t, and just ignored the negative sign, rather than considering the next positive solution. Another common error lay in using the wrong calculator mode once more.

Q15.

Some candidates gave a very concise and correct solution to this differential equation. Most candidates made some attempt to separate the variables and integrate, although it was not always clear that they had done this, with a stray x sometimes remaining on the left hand side. There were many errors in the integration, the common ones being ln y, resulting from poor algebra in separation, and a coefficient error in the integral of cos $3x$, the 3 itself often being dropped and the integral given as sin x.

Most candidates did include an arbitrary constant and tried to find it using the given conditions, although a few just substituted these into their solution without a + C, or directly into the differential equation, and just produced nonsense.

Q16.

Nearly all candidates successfully found the partial fractions in part (a), most substituting x = 1 and x = –1; although some used other values or equated coefficients.

In part (b) most candidates then went on to use their partial fractions to integrate the given expression correctly. The error that was sometimes seen was to integrate $(x - 1)^{-1}$ to $(x - 1)^0$, and similarly for $(x + 1)^{-1}$.

Most candidates could make a start to part (c) of the question, but many rearranged the 2 and 3, and this detracted from the realisation that with y and 3 taken to the other side of the equation, they had the integral they had just done in part (b). Some who did realise this, proceeded to give an incorrect integral because they had "lost" the 2. Most candidates knew they needed to separate the variables but for many candidates the algebra and ensuing attempts at integration were full of errors. Turning the denominators y and $(x^2 - 1)$ into numerators was common, as was integrating $(x^2 - 1)^{-1}$ to an expression involving ln $(x^2 - 1)$.

Those who had taken the 2 to the left hand side, often got confused with the integral of $(2y)^{-1}$, this becoming ln $2y$. The problem for most candidates seemed to be that they associated the 2 in the question with the 2 in the answer, although the 2 in the answer came from the given values of x and y. Most candidates did include an arbitrary constant and tried to use the given values to find it, often claiming, after some dubious manipulation, that they had derived the given result. Few fully correct answers were seen, but where they were, it was reassuring to see confident use of the log laws in deriving the result.

Q17.

The responses to this question were very mixed, varying from the very poor to some excellent answers. Most, but not all, candidates did attempt to separate and integrate in part (a)(i) with some making an error in their integrals, and rather more omitting a constant. Such candidates could make no further progress in part (a)(ii) without a constant to find. Some candidates thought to bring e^{-1} into their answer in part (a)(i), instead of the expected conventional constant, and got themselves confused in part (a)(ii) as they could not convincingly obtain the given answer. Some candidates made the anticipated error at the exponentiation stage of $\ln y = -\cos t + c \Rightarrow y = e^{-\cos t} + e^{c}$. Many candidates could not justify the 50, other than saying it was the initial value.

For part (b) many candidates realised they had been given the required expression in part (a) and substituted and evaluated correctly, although some ignored the request to answer to the nearest centimetre, any greater accuracy being rather meaningless in this modelling problem. Some used degrees instead of radians. Other candidates substituted into the differential equation. In part (b)(ii), although most candidates knew in principle what to do, they could not find the second derivative correctly. Many treated y as a constant or as if it were t, rather than using implicit differentiation.

Some candidates did attempt to differentiate the given answer in part (a)(ii) twice, but usually made an error in their use of the chain rule. Some candidates did get the differentiation correct, but failed to complete their argument by demonstrating that the first derivative was zero at $t = \pi$, so confirming this point as a turning point.

Q18.

For part (a), most candidates knew that they were supposed to separate the variables and integrate, although interpretations of this varied widely. There were some impressive answers with clear integration leading to the final requested result. In contrast, some candidates were not sure what to do with dy and dx, these sometimes appearing as denominators. The weakest candidates simply substituted the given values of x and y into the differential equation and produced meaningless nonsense. Most candidates who had separated correctly at least got as far as $k\sqrt{1+2y}$ for their y integral, although an index of $\frac{3}{2}$ or $-\frac{1}{2}$ was a fairly common error.

It was pleasing to see the integration of $\frac{1}{x^2}$ done correctly by many candidates, with the common error being in the index. It was also pleasing that most candidates included an arbitrary constant in their solution and tried to find it using the given condition.

In part (b), most candidates who had $k\sqrt{1+2y}$ in their solution realised that they were to square it to move towards the given result, but many attempts were poor, with $(\mathrm{f}(x)+c)^2 = (\mathrm{f}(x))^2 + c$ being frequently seen, despite there being a term in $\frac{1}{x}$ in the given solution. Some candidates who had correctly introduced a constant c in part (a) proceeded to manipulate their solution without including c in the manipulation, and thus failed to score the mark for finding it.

Q19.

Most knew in principle that they were to separate the variables and integrate, but there was a lot of mishandling of the $(x-6)^{\frac{1}{2}}$ term. Most candidates integrated dt correctly, and included a constant, and so were able to show they could use the initial conditions to find a value for the constant even if their x term was incorrect. Common errors in the integration of the x term were in the coefficient of$(x-6)^{\frac{1}{2}}$ or multiples of $(x-6)^{\frac{3}{2}}$ or $\ln(x-6)^{\frac{1}{2}}$.

Some candidates who had done all the integration correctly failed to give their final answer in the form requested.

For part (b) most candidates showed they understood the context through making a sensible comment in part (i), although a few thought it meant the tank was empty.

Those who just stated $\frac{dx}{dt}$ = 0 gained no credit.

In part (b) (ii) quite a few candidates lost the marks because they used x = 48 (70–22) rather than x = 22. However, there were many candidates who gave good full answers to this question even if a few lost sight of the correct units for the final answer. This was condoned.

Q20.

Many full mark answers were seen to this question with candidates demonstrating confidence in solving the equation. Some lost the last mark, or two, through incorrect manipulation to the requested j = f(x)form. Finding the value of the constant after inverting to $y = -\frac{1}{3x^2}$ was the common error. Poorer attempts showed no knowledge of the separation of variables technique, and candidates attempted to use integration by parts. Others made errors in the algebra, or in the integration or did not include a constant. Integrating $\frac{1}{y^2}$ to an expression involving a ln function was fairly common.

Q21.

Many candidates gained full marks on this question presenting clear, good quality answers.

Part (b)(i) Most candidates knew they were required to separate and integrate and attempted to do so, sometimes with poor notation and incorrect integrals but their intention was clear. Most candidates also included a constant of integration and attempted to find it using the given initial conditions. Some neglected to express t in terms of x with some going straight to the result given in part (b)(ii).

Part (b)(ii) This was often done well but the common error was in taking exponentials, with $e^{\frac{t}{40}} = e^{\ln 70} + e^{\ln(x-15)}$ being fairly common.

Q22.

This question was usually answered well. However some students forgot the minus sign in the initial equation 12 $\frac{dv}{dt} = -4v^{\frac{1}{3}}$ whilst others had difficulty with the evaluation of $\int v^{-\frac{1}{3}} dv$.

Most students attempted to find the '+c' term and obtained the printed result. Nearly all students correctly found the time when the particle came to rest.

Q23.

Most students answered part (a) correctly but some did not replace their a by $\frac{dv}{dt}$. In part

(b) students were expected to use $\int \frac{1}{100-v}\,\mathrm{d}v = -\int \frac{1}{40}\,\mathrm{d}t$; some incorrectly split $\int \frac{1}{100-v}\,\mathrm{d}v$ into $\int \frac{1}{100} - \frac{1}{v}\,\mathrm{d}v$, whilst others incorrectly used $\int (100 - v)\mathrm{d}v = \int \frac{1}{40}\,\mathrm{d}t$.

A common error was in forgetting the minus sign in the evaluation of $\int \frac{1}{100-v}\,\mathrm{d}v$. Most students attempted to find the '+ c' term in the solution of the differential equation.

Q24.

This question was answered well by most candidates. However, in part (a), a few candidates did not show the required step $0.4\frac{\mathrm{d}v}{\mathrm{d}t} = 2 - 4v$; others lost and recreated the minus signs, eg $\frac{\mathrm{d}v}{\mathrm{d}t} = 5 - 10v = -10(0.5 + v) = -10(v - 0.5)$, which was penalised. In part (b), the majority of candidates used $\int \frac{1}{v-0.5}\,\mathrm{d}v = -\int 10\mathrm{d}t$ and then solved this by integrating both sides.

The integration $\int \frac{1}{v-0.5}\,\mathrm{d}v$ caused some candidates difficulty, finding terms such as $\frac{1}{\frac{1}{2}v^2 - 0.5v}$ etc. The use of logarithms was often incorrect, with $\ln(v - 0.5) = -10t + \ln 0.5$ commonly becoming $v - 0.5 = e - 10t + 0.5$

Q25.

Most candidates answered part (a) correctly but some just wrote down the printed result.

In part (a) some candidates started with $m\frac{\mathrm{d}v}{\mathrm{d}t}\mathrm{d}v = -9.8v$ but found that this did not give the printed result, and just invented a term in g. This was not accepted. Some were penalised for ignoring the m term or for changing signs.

In part (b) candidates were expected to use $\int \frac{1}{v-5}\,\mathrm{d}v = -1.96\mathrm{d}t$; some split $\int \frac{1}{v-5}\,\mathrm{d}v$ into $\int \frac{1}{v} - \frac{1}{5}\,\mathrm{d}v$. However there were many good answers to this part with both the integrations required being successfully completed. The difficulty which some candidates had was in the simplification needed to convert $\ln(v - 5) = -1.96t + \ln 2$ into $v - 5 = 2e^{-1.96t}$ which often became $v - 5 = e^{-1.96t} + 2$

Q26.

Most candidates answered part (a)(i) correctly but some just wrote down the printed result.
In part (a)(ii), candidates needed to start from $F = ma$ and show the cancellation of the mass (65). Some were penalised for ignoring the m term or ignoring the minus sign.

In part (b), candidates were asked to show that $\int \frac{1}{v - 2.45} \mathrm{d}v = -\int 4 \mathrm{d}t$ many simply used this equation and were penalised. There were, however, many good answers to this part, with both the integrations required being successfully completed. The difficulty which candidates had was in the simplification needed to convert $\ln(v - 2.45) = -4t + \ln 17.15$ into
$v - 2.45 = 17.15e^{-4t}$, which often became $v - 2.45 = 17.15 + e^{-4t}$.

Q27.

Most candidates started from $F = ma$ and answered part (a) correctly. Others were penalised for ignoring the m term or ignoring the minus sign. There were many good answers to part (b), although a significant number of candidates used 'inventive' algebra. A few candidates who integrated to find the equation in v and t forgot the $+c$ term.

Q28.

Most candidates started from $F = ma$ and answered part (a) correctly. Others were penalised for ignoring the m term or ignoring the minus sign. There were many good answers to part (b), although some used inventive algebra. The solution to part (c) was usually correct, although candidates often forgot to obtain ± 1, for the square root of v, when they found the square-root of the equation obtained in part (b).

In part (d) candidates who integrated v to find the distance travelled often forgot the + d term which was required to be shown to be equal to zero.

Those who used integration with limits commonly used $\int_{16}^{1} (4 - 0.1t)^2 \mathrm{d}t$ forgetting that the integral was with respect to t so that $\int_{0}^{30} (4 - 0.1t)^2 \mathrm{d}t$ was required. A considerable number of candidates, in part (d), tried to use equations for motion with constant acceleration.

Q29.

The majority of candidates used $\frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{\lambda}{v^{\frac{1}{4}}}$ and then solved this by using

$\int v^{\frac{1}{4}} \mathrm{d}v = \int -\lambda \mathrm{d}t$. Some used $a = \frac{\mathrm{d}v}{\mathrm{d}t}$, together with $v = u + at$, so that $v = u - \frac{\lambda}{v^{\frac{1}{4}}} t$ was often seen.

Many candidates found $\frac{4}{5} v^{\frac{5}{4}} = -\lambda t + \frac{4}{5} y^{\frac{5}{4}}$, but the solution for v was not often found correctly.

A common result was "$v^{\frac{5}{4}} = u^{\frac{5}{4}} - \frac{5}{4}\lambda t \Rightarrow v = u - \left(\frac{5}{4}\lambda t\right)^{\frac{4}{5}}$".

Q30.

Most candidates started from $F = ma$ and answered part (a) correctly. Others were penalised for starting with $\frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{0.08v^2}{0.05}$. There were many good answers to part (b), with very few candidates being unable to obtain $v = \frac{15}{5 + 24t}$ from $\frac{1}{v} = 1.6t + \frac{1}{3}$.

Q31.

The answer printed in part (a) enabled virtually all candidates to create their solution. However, a considerable minority were not convincing. Part (b) was well answered, with most candidates appreciating that the differential equation needed to be solved by separating variables. Some found the resulting integral, $\int v^{-\frac{3}{2}}\,\mathrm{d}v$, challenging, and the $+c$ term was often omitted. In part (c), rather than using $\frac{2}{\sqrt{v}} = \lambda t + \frac{2}{3}$ to obtain t when $v = 4$, many used the last result, $v = \frac{36}{(2 + 3\lambda t)^2}$ when $v = 4$, and forgot to eliminate the negative root when calculating the square root of $4 = \frac{36}{(2 + 3\lambda t)^2}$.

Q32.

Parts (a), (b) and (c) were answered well. In part (d), few candidates used the fact that the work done by the force was equal to the change in kinetic energy. Most used complicated methods which rarely gave an appropriate answer.

Q33.

Virtually all candidates obtained $\frac{\mathrm{d}v}{\mathrm{d}t} = -0.05v$: only a few ignored the required step $m\frac{\mathrm{d}v}{\mathrm{d}t} = -0.05mv$. The equation $\int \frac{\mathrm{d}v}{v} = -\int 0.05\ \mathrm{d}t$ was a necessary step which needed to be seen in part (b). Candidates knew roughly how to obtain $v = 20e^{-0.05t}$. Unfortunately, too often algebraic skills were not sufficient and the equation
$\ln v = -0.05 + c$ regularly became $v = e^{-0.05t} + c$ before becoming $v = Ae^{-0.05t}$. This and similar errors were not condoned.

Q34.

A significant number of candidates created $\frac{dv}{dt} = -\lambda v$ without proper justification. The equation $-\lambda mv = ma$ was a necessary step which needed to be seen. Even more knew roughly how to obtain $v = Ue^{-\lambda t}$, but too often algebraic skills were not sufficient and $\ln v = -\lambda t + \ln u$ regularly became $v = e^{-\lambda t} + U$ before becoming $v = Ue^{-\lambda t}$. This and similar errors were not condoned.

Q35.

This question required the use of a differential equation, but some candidates tried to use constant acceleration equations. There were a number of good, complete solutions. Some candidates were on the right track with their solutions, but made algebraic or integration errors. A few candidates omitted the negative sign when they started the question and ended up with a final answer which simply had a missing negative sign.